

Power-Law Scaling Boundary Release for HAOS-IIP

Tomislav RupiĆ
Independent Researcher

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Abstract

This paper freezes release 53.3 of HAOS-IIP as a power-law scaling boundary result in the external physics-bridge layer. The previous bridge table contained a broad power-law scaling **OPEN** row because the raw local-gradient read had minimum power-fit $R^2 = 0.321965$, far below the threshold $R^2 \geq 0.95$. Release 53.3 keeps that raw failure explicit, but separates it from the already-frozen shell-native, law-aware reconstruction. The raw local-gradient read remains open with minimum $R^2 = 0.3220$, maximum $|slope + 2| = 1.2651$, scaled-flux CV 0.3139, and profile drift 0.5249. The shell-native read passes as a bounded continuum-like power-law proxy with minimum power-fit $R^2 = 0.9997$, maximum $|slope + 2| = 0.0034$, scaled-response CV 0.0070, and profile drift 0.0089. The correct conclusion is therefore a reconstruction split: raw local-gradient power scaling remains open, while shell-native power-law scaling closes on the tested scalar-carrier window.

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1 Scope

Release 53.3 is an external audit over frozen scalar-carrier artifacts.

It does not modify HAOS core, alter the scalar operator, introduce new dynamics, or claim a continuum limit. It only asks:

does the power-law scaling proxy fail everywhere, or only in the raw local-gradient readout?

2 Source Artifacts

The audit uses:

- raw local-gradient recoverability artifact: `data/scalar_kernel_graph_recoverability_gradient_latest.json`;
- shell-native recoverability artifact: `data/scalar_kernel_graph_recoverability_gradient_shell_native_latest.json`;
- audit runner: `numerics/simulations/scalar_kernel_graph_power_law_scaling.py`.

No scalar simulation is redefined by this release. The audit compares existing frozen measurements.

3 Measured Quantities

For each readout the release records:

- minimum power-fit R^2 ;
- maximum $|slope + 2|$;
- maximum scaled-response coefficient of variation;
- maximum refinement-profile drift.

The shell-native read also inherits the validated Green-law fit guardrail from the 52.2 reconstruction.

4 Frozen Result

Readout	Status	Min R^2	Max $ slope + 2 $	Max CV	Max drift
raw local gradient	OPEN	0.3220	1.2651	0.3139	0.5249
shell native	PASS	0.9997	0.0034	0.0070	0.0089

Table 1: Power-law scaling split across raw and shell-native readouts.

The weakest raw case is:

- `disorder|n=11|jitter=0.020`;
- slope -0.7349 ;
- power-fit $R^2 = 0.3220$;
- scaled-flux CV 0.2880.

The weakest shell-native case is:

- `disorder|n=13|jitter=0.040`;
- Green fit $R^2 = 0.9856$;
- slope -1.9966 ;
- power-fit $R^2 = 0.9998$;
- scaled-response CV 0.0067.

5 Interpretation

The old broad bridge statement was:

power-law scaling remains open because raw power-fit R^2 is too low.

The updated statement is:

raw local-gradient power scaling remains open, but shell-native law-aware power scaling closes as a bounded continuum-like proxy on the tested scalar-carrier window.

This matters because the failure mode is now localized. The carrier does not support a raw local derivative force-law readout, but it does support the shell-native reconstruction aligned with the validated Green structure.

6 Non-Claims

This release does not claim:

- a general continuum limit;
- continuum field equations;
- exact force-law ontology;
- raw local vector-force closure;
- broad substrate universality;
- equivalence to GR, QFT, or quantum gravity.

7 Artifacts

- script: `numerics/simulations/scalar_kernel_graph_power_law_scaling.py`;
- operator note: `experiments/operators/Scalar_Kernel_Graph_Power_Law_Scaling_v1.md`;
- frozen result: `data/20260423_183612_scalar_kernel_graph_power_law_scaling.json`;
- latest result: `data/scalar_kernel_graph_power_law_scaling_latest.json`;
- summary plot: `plots/20260423_183612_scalar_kernel_graph_power_law_scaling_summary.png`;
- profile plot: `plots/20260423_183612_scalar_kernel_graph_power_law_scaling_profiles.png`.

8 Next Boundary

The next honest step is to add a larger refinement point where runtime permits, then test whether the shell-native power-law scaling result survives one additional kernel-graph family before using stronger universality language.

9 Conclusion

Release 53.3 turns the power-law scaling item from an undifferentiated open row into a precise reconstruction boundary. Raw local-gradient scaling remains open. Shell-native scaling passes. That is the correct stopping point: a bounded continuum-like proxy under a stated reconstruction, not a continuum-limit proof.